

Advanced Algebra



Simplifying rational expressions

- 1 Simplify $\frac{x^2 - 4}{x^2 + x - 6}$.
 - 2 Simplify $\frac{x^2 - 5x + 6}{x^2 - 9}$.
 - 3 Combine into a single fraction: $\frac{2}{x+1} + \frac{3}{x-1}$.
 - 4 Simplify $\frac{3x^2 - 12}{x^2 - 4x + 4}$.
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Absolute value equations

- 5 Solve $|2x - 5| = 9$.
 - 6 Solve $|3x + 1| = |x - 5|$.
 - 7 Solve $|x - 4| + 3 = 10$.
 - 8 Solve $|4x + 3| = 11$.
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Polynomial division, remainder & factor theorems

- 9 Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.
 - 10 Given $f(x) = x^3 + 3x^2 - 4x - 12$:
 - a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.
 - b Hence factor $f(x)$ completely.
 - 11 Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.
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Logarithmic equations

- 12 Solve $\log_3(x + 5) = 2$.
- 13 Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.

14 Solve $2\log_5 x = \log_5 16$.

15 Solve $\log_3 (x - 2) = 4$.

Exponential equations

16 Solve $3^{2x-1} = 27$.

17 Solve $5^{x+2} = 5^{3x-4}$.

18 Solve $2^x = 15$ for x , giving your answer to three decimal places.

19 Solve $4^x = 2^{x+3}$.

Nonlinear systems of equations

20 Solve the system $y = x^2 - 1$ and $y = 2x + 2$.

21 Solve the system $x^2 + y^2 = 25$ and $y = x + 1$.

22 Solve the system $y = x^2 - 4x + 3$ and $y = -x + 3$.

Rational equations and extraneous solutions

23 Solve $\frac{4}{x-1} = \frac{2}{x+1}$.

24 Solve $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

25 Solve $\frac{3}{x+4} = \frac{1}{x-2}$.

Systems of inequalities and linear programming

26 A feasible region is bounded by the constraints $x \geq 0$, $y \geq 0$, $x + y \leq 6$, and $2x + y \leq 10$. Find the vertices of the feasible region.

27 Using the feasible region from the previous question, find the maximum value of $P = 3x + 2y$ over the region, and state where it occurs.

A well-designed word problem: production planning at a furniture workshop

- 28** This question has three parts. A furniture workshop makes tables and chairs. Each table requires 4 hours of carpentry and 2 hours of finishing; each chair requires 1 hour of carpentry and 1 hour of finishing. Each week the shop has at most 32 hours of carpentry time and 20 hours of finishing time available. Each table earns a profit of 70 dollars and each chair earns a profit of 30 dollars. Let x be the number of tables and y the number of chairs produced in a week. (a) Write the system of inequalities describing the constraints (including $x \geq 0$ and $y \geq 0$). (b) Find the vertices of the feasible region. (c) Determine the number of tables and chairs that maximizes the weekly profit $P = 70x + 30y$, and find the maximum profit.
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Multiple choice — easy

- 29** Solve $2^{2x-1} = 128$.

a. 5
b. 7
c. 3
d. 4

- 30** Solve $3^{2x-1} = 3$.

a. 3
b. 0
c. 2
d. 1

- 31** Solve $2^{2x-1} = 2$.

a. 0
b. 1
c. 3
d. 2

32 Solve $2^{2x-1} = 2048$.

- a. 7
- b. 6
- c. 5
- d. 11

33 Solve $2^{2x-1} = 32$.

- a. 5
- b. 3
- c. 2
- d. 4

34 Solve $3^{2x-1} = 19683$.

- a. 4
- b. 5
- c. 9
- d. 6

35 Solve $3^{2x-1} = 2187$.

- a. 3
- b. 4
- c. 7
- d. 5

36 Solve $3^{2x-1} = 27$.

a. 2

b. 4

c. 3

d. 1

37 Solve $2^{2x-1} = 512$.

a. 6

b. 4

c. 9

d. 5

38 Solve $7^{2x-1} = 16807$.

a. 4

b. 2

c. 5

d. 3

39 Solve $7^{2x-1} = 1977326743$.

a. 6

b. 11

c. 7

d. 5

40 Solve $7^{2x-1} = 40353607$.

a. 4

b. 6

c. 9

d. 5

41 Solve $5^{2x-1} = 48828125$.

a. 11

b. 5

c. 6

d. 7

42 Solve $3^{2x-1} = 243$.

a. 2

b. 3

c. 5

d. 4

43 Solve $5^{2x-1} = 125$.

a. 1

b. 3

c. 4

d. 2

Multiple choice — medium

- 44 Use the remainder theorem to find the remainder when $f(x) = x^3 - 5x^2 + 4x$ is divided by $(x - 0)$.
- a. -2
 - b. 3
 - c. 2
 - d. 0
- 45 Use the remainder theorem to find the remainder when $f(x) = x^3 + 5x^2 - x - 5$ is divided by $(x - 1)$.
- a. 0
 - b. -5
 - c. -2
 - d. 2
- 46 Use the remainder theorem to find the remainder when $f(x) = x^3 - 2x^2 - 5x - 1$ is divided by $(x - (-4))$.
- a. -75
 - b. -79
 - c. -45
 - d. -77
- 47 Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 - 2x + 2$ is divided by $(x - (-4))$.
- a. -70
 - b. -54
 - c. -72
 - d. -68

- 48 Use the remainder theorem to find the remainder when $f(x) = x^3 + 6x^2 + 9x - 7$ is divided by $(x - (-5))$.
- a. -29
 - b. -25
 - c. -177
 - d. -27
- 49 Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 3x - 5$ is divided by $(x - (-4))$.
- a. -27
 - b. -23
 - c. -57
 - d. -25
- 50 Use the remainder theorem to find the remainder when $f(x) = x^3 + x^2 - 7x + 2$ is divided by $(x - (-4))$.
- a. -16
 - b. -18
 - c. -34
 - d. -20
- 51 Use the remainder theorem to find the remainder when $f(x) = x^3 + 5x^2 + 9$ is divided by $(x - 2)$.
- a. 37
 - b. 35
 - c. 17
 - d. 39

- 52 Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 + 7x - 9$ is divided by $(x - 5)$.
- a. 151
 - b. 128
 - c. 126
 - d. 124
- 53 Use the remainder theorem to find the remainder when $f(x) = x^3 - x^2 - 4x - 8$ is divided by $(x - (-2))$.
- a. -14
 - b. -12
 - c. -10
 - d. -8
- 54 Use the remainder theorem to find the remainder when $f(x) = x^3 - 4x - 4$ is divided by $(x - (-1))$.
- a. 1
 - b. 2
 - c. -3
 - d. -1
- 55 Use the remainder theorem to find the remainder when $f(x) = x^3 + 6x^2 - 4x - 7$ is divided by $(x - (-5))$.
- a. 40
 - b. 38
 - c. -112
 - d. 36

- 56 Use the remainder theorem to find the remainder when $f(x) = x^3 - 3x^2 + 6x + 1$ is divided by $(x - (-1))$.
- a. -7
 - b. -9
 - c. -11
 - d. -6
- 57 Use the remainder theorem to find the remainder when $f(x) = x^3 - 5x^2 + x - 4$ is divided by $(x - (-5))$.
- a. -259
 - b. -261
 - c. -134
 - d. -257
- 58 Use the remainder theorem to find the remainder when $f(x) = x^3 + 4x^2 + x + 7$ is divided by $(x - 0)$.
- a. 10
 - b. 7
 - c. 9
 - d. 5

Multiple choice — hard

- 59 Solve $\log_2(x + 3) + \log_2(x - 3) = 4$.
- a. 7
 - b. 3
 - c. -5
 - d. 5

60 Solve $\log_2(x + 2) + \log_2(x - 2) = 5$.

a. 6

b. 8

c. -6

d. 4

61 Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.

a. -3

b. 5

c. 3

d. 1

62 Solve $\log_3(x + 3) + \log_3(x - 3) = 3$.

a. 4

b. 6

c. -6

d. 8

63 Solve $\log_3(x + 4) + \log_3(x - 4) = 2$.

a. 7

b. -5

c. 3

d. 5

64 Solve $3^x = 189$ for x , giving your answer to three decimal places.

a. 5.771

b. 5.271

c. 4.271

d. 4.771

65 Solve $5^x = 141$ for x , giving your answer to three decimal places.

a. 2.575

b. 4.075

c. 3.075

d. 3.575

66 Solve $7^x = 361$ for x , giving your answer to three decimal places.

a. 4.026

b. 2.526

c. 3.026

d. 3.526

67 Solve $5^x = 372$ for x , giving your answer to three decimal places.

a. 3.178

b. 4.178

c. 4.678

d. 3.678

68 Solve $5^x = 410$ for x , giving your answer to three decimal places.

a. 3.738

b. 4.238

c. 4.738

d. 3.238

69 Solve the system $y = x^2 + 4x + 1$ and $y = -5x - 7$. Find the x -coordinates of the intersection points.

a. $-8, -3$

b. $-8, -1$

c. $-6, -1$

d. $8, 1$

70 Solve the system $y = x^2 + x + 10$ and $y = -5x + 10$. Find the x -coordinates of the intersection points.

a. $-4, 0$

b. $6, 0$

c. $-6, 0$

d. $-6, -2$

71 Solve the system $y = x^2 + 17x + 47$ and $y = 2x - 9$. Find the x -coordinates of the intersection points.

a. $-8, -9$

b. $-8, -7$

c. $8, 7$

d. $-6, -7$

72 Solve the system $y = x^2 + 13x + 27$ and $y = 4x + 7$. Find the x -coordinates of the intersection points.

a. $-5, -4$

b. $5, 4$

c. $-3, -4$

d. $-5, -6$

73 Solve the system $y = x^2 - 8x + 16$ and $y = 3x - 8$. Find the x -coordinates of the intersection points.

a. $3, 8$

b. $5, 8$

c. $3, 6$

d. $-3, -8$